	x_1	x_2
S_1	1	1
S_2	1	2
S_3	2	2
S_4	8	8
S_5	8	9
S_6	9	8

$K = 2$

C

Step 1: Randomly assign cluster to observation

C_1	C_2
S1	S2
S3	S5
S4	S6

Cluster 1 :-

(1,1)
(2,2)
(8,8)

Cluster 2

(1,2)
(8,9)
(9,8)

Step 2: Compute the centroid for all clusters.

Centroid C_1 $= \left(\frac{11}{3}, \frac{11}{3}\right)$

$(3.66, 3.66)$

Centroid C_2 $= \left(\frac{18}{3}, \frac{19}{3}\right)$

$(6, 6.33)$

Step 3: Process each point & assign the cluster based on the min distance to the centroid

S_1 (1,1) with C_1 $\rightarrow \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$\Rightarrow \sqrt{(3.66-1)^2+(3.66-1)^2}$

$= 3.76$

S_1 (1,1) with C_2 $\Rightarrow \sqrt{(6-1)^2+(6.33-1)^2}$

$= 7.3$

S_3 (2,2) with C_1 $\Rightarrow \sqrt{(3.66-2)^2+(3.66-2)^2}$

$= 2.34$

S_3 (2,2) with C_2 $\Rightarrow \sqrt{(6-2)^2+(6.33-2)^2} = 5.894$

Date

S_4 $(8,8)$ with C_1 $\Rightarrow \sqrt{(3.66-8)^2+(3.66-8)^2}$

$= 6.137$

S_4 $(8,8)$ with C_2 $\Rightarrow \sqrt{(6-8)^2+(6.3-8)^2}$

$= 2.624$

S_2 $(1,2)$ with C_1 $\Rightarrow \sqrt{(3.66-1)^2+(3.66-2)^2}$

$= 3.13$

S_2 $(1,2)$ with C_2 $\Rightarrow \sqrt{(6-1)^2+(6.3-2)^2}$

$= 6.59$

S_5 $(8,9)$ with C_2 $\Rightarrow \sqrt{(6-8)^2+(6.3-9)^2}$

$\Rightarrow 3.36$

S_5 $(8,9)$ with C_1 $\Rightarrow \sqrt{(3.66-8)^2+(3.66-9)^2}$

$= 6.881$

S_6 $(9,8)$ with C_1 $\Rightarrow \sqrt{(3.66-9)^2+(3.66-8)^2}$

$\Rightarrow 6.881$

S_6 with C_2 $\Rightarrow \sqrt{(6-9)^2+(6.3-8)^2}$

$= 3.44$

	with C_1	with C_2
S1	3.76	7.3
S2	3.13	6.59
S3	2.34	5.894
S4	6.137	2.624
S5	6.881	3.36
S6	6.881	3.44

∴ New cluster

⇒ S1, S2, S3 | S4, S5, S6

$C_1 \Rightarrow (1,1)$, $(1,2)$, $(2,2)$

$C_2 \Rightarrow (8,8)$, $(8,9)$, $(9,8)$.

Centroid $(C_1) = \left(\frac{4}{3}, \frac{5}{3}\right)$

$= (1.33, 1.66)$

$C_2 = \left(\frac{25}{3}, \frac{25}{3}\right)$

$(8.33, 8.33)$

$S_1(1,1)$ with C_1 $\Rightarrow \sqrt{(1.33-1)^2 + (1.66-1)^2}$

$= 0.737$

$S_1(1,1)$ with C_2 $= \sqrt{(8.33-1)^2 + (8.33-1)^2}$

$= 10.36$

$S_2(1,2)$ with C_1 $= \sqrt{(1.33-1)^2 + (1.66-2)^2}$

$= 0.473$

$S_2(1,2)$ with C_2 $= \sqrt{(8.33-1)^2 + (8.33-2)^2}$

$= 9.68$

$S_3(2,2)$ with C_1 $= \sqrt{(1.33-2)^2 + (1.66-2)^2}$

$= 0.751$

$S_3(2,2)$ with C_2 $= \sqrt{(8.33-2)^2 + (8.33-2)^2} = 8.95$

$S_4(8,8)$ with C_1 $= \sqrt{(1.33-8)^2 + (1.66-8)^2} = 9.202$

$S_4(8,8)$ with C_2 $= \sqrt{(8.33-8)^2 + (8.33-8)^2} = 0.46$

$S_5(8,9)$ with C_1 $= \sqrt{(1.33-8)^2 + (1.66-9)^2} = 9.917$

$S_5(8,9)$ with C_2 $= \sqrt{(8.33-8)^2 + (8.33-9)^2} = 0.74$

$S_6(9,8)$ with C_1 $= \sqrt{(1.33-9)^2 + (1.66-8)^2} = 9.95$

$S_6(9,8)$ with C_2 $= \sqrt{(8.33-9)^2 + (8.33-8)^2} = 0.74$

	with C_1	with C_2
S_1	0.737	10.36
S_2	0.473	9.68
S_3	0.751	8.95
S_4	9.202	0.46
S_5	9.917	0.74
S_6	9.95	0.74

$\therefore$ The clusters remains the same